

Test Project

Pre version of Test Project

Autonomous Mobile Robotics

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1. Introduction

The primary focus of the WorldSkills Shanghai 2026 Autonomous Mobile Robotics Pre-Test Project is to outline the requirements for Competitors to assemble, prepare, manage, and repair robots to interpret multiple environments and tasks introduced at the competition. It is expected that robots will use the environmental information they have gathered to manage both their overall mobility and their target-object-handling experiences.

As part of the WorldSkills Shanghai 2026 Competition, Competitors are tasked with constructing, programming, and operating an autonomous mobile robot that meets all the performance requirements.

In Lyon, we introduced a complete overhaul of the entire skill to focus more on the skills themselves. For WorldSkills Shanghai 2026, we continue this change and improve upon it, which means more simplification and module combination. For example, instead of testing vision in an isolated environment outside the robot environment, Competitors will be using vision in the robot environment.

2. Description of project and tasks

2.1 Main Test Project

Teams of two Competitors are required to develop and assemble an autonomous mobile robot that efficiently operates in custom environments that represent real-world scenarios or tasks.

The main Test Project is broken down into various modules in accordance with the WorldSkills Occupation Standards (WSOS). Competitors are expected to identify the primary set of autonomous mobile robot performance requirements through analysis of the information provided in this document:

1. Robots are required to move mostly autonomously throughout any performance evaluation environment.
2. Robots are required to move objects within the performance environment.
3. Robots are required to take control of the various target objects within the performance environment, and or interact with sensors and vision systems.

Note: The robot base will be assembled on C1. Competitors can work on OMS assembly on C-2. However, it will not be attached until the afternoon of C1.

3. Instructions to the Competitor

3.1 Competition Requirement

3.1.1 Competitor Parts

Competitors will receive one WorldSkills 2026 Shanghai Full Autonomous Mobile Robotics Component Collection from the sponsor Studica. Any of the 4 different drive bases: Mecanum, Differential, Differential (All-Terrain), and X-Drive, can be built from the collection. Also included are the electronics and extra components. A roll of the dice, the morning of C1, in front of all Competitors and

Experts, will decide the drive base to use for the rest of the Competition's Days. All Competitors will be **required** to use the same drive base. Drive bases are assessed at C1 lunch. **Drive bases may not be modified in any way from the original design.** Adding items such as sensors to the drive base is allowed.

Outside the collection, Competitors are given a 1,000-point budget. Budgets are a major part of any technician or engineer's job during a project. Having a budget will test how the Competitors balance performance and cost. Previous competitions had a limit put on 3D-printed or polycarbonate components based on weight and size. For WorldSkills Shanghai 2026, the 3D-Printed and sheet components will be taken from the Competitor budget. 3D printed parts will be priced at 0.5 points/g. Sheet components will be priced at 0.25 points/g. Any product on <https://www.studica.co/> can be used. The budget cost is marked on each component page. Competitors will be required to use the budget for parts to achieve the best results.

NOTE: ALL COMPETITOR SPARE PARTS (3D-PRINTED, SHEET, OTHERS) WILL BE STORED IN THE SCM ROOM. WHEN A SPARE PART IS REQUIRED, IT WILL BE CHECKED BY AN EXPERT AND THAT SPARE PART WILL BE ADDED TO THE BUDGET OF THOSE COMPETITORS.

In previous Competitions, there was a size limit to robots. The size limit is now gone, as Competitors will all share the same dive base, which will be guaranteed to fit in the court.

3.1.2 Replacement Parts

Damaged parts that need replacement will be replaced by the appropriate sponsor. Parts with legitimate defects will be replaced at no cost to the Competitor. **Parts deemed damaged due to the Competitor's misuse will be replaced and added to the Competitor's budget.**

3.2 Activities to be completed before the Competition

Competitors are required to prepare well in advance. This would include practicing building and wiring all four drive bases. Knowing and preparing code to operate all four drive bases.

Preparing an itemized, accurate budget to hand in to the experts. Experts will verify everything in the budget and its accuracy.

Cable harnesses will no longer be allowed into the competition. Competitors will be required to manufacture some or all of the cables at the competition. Appropriate cable, crimps and crimpers will be provided to Competitors. The Competitors must be skilled at selecting the appropriate cables and connectors for each action on the robot. Example: using 10–16 A.W.G. wire for motors.

4. Equipment, Machinery, Installations, and Materials Required

4.1 Equipment Provided by Competition Organizers

- A WorldSkills Shanghai 2026 Full Autonomous Mobile Robotics Collection will be provided on C-2.
- All Competitors will receive a laptop on C-2 with the core programming tools installed. (two laptops per Member)
- All equipment, tooling, and materials are in the Infrastructure List – see WorldSkills Discussion Forum and Technical Description.
- All materials for any surprise Test Project as listed in the Technical Description.

- Testing equipment for the fault-finding module.
- Objects required for manipulation during the performance tasks and testing.
- Test courts for the performance evaluation tasks.
- Parts and tools required to complete the prototyping module and maintain the robot throughout the competition.

4.1.1 VMX

The VMX is the only controller allowed on the Competitor robots. ***Note: Competitor laptops are allowed to process information and send it back to the VMX. However, it is at the Competitor's own risk if the wireless connection drops.**

Competitors may use any software that they can get running on the VMX. Note that for some software, the competition organizers may require you to show a valid licence, for example, LabVIEW.

4.1.2 Competitor Computer

Laptops will be provided for each team at the competition. The specifications are in the infrastructure list. All laptops will be dual-boot with Windows and Linux. For all modules, Competitors may use any software, understanding that no support will be given for software not provided by the Sponsor. For some software loaded by the Competitor on C-2, proof of a valid licence may be required.

The Competitor computer will contain all required software for the competition; some software will be on Windows and others on Ubuntu:

- Dual Boot Windows and Linux (Ubuntu)
- WPILib Java and C++
- Local version of Docs Page
- Control Station Console
- Studica Hardware Manager
- VNC Viewer or equivalent
- Putty/Terra Term/CoolTerm
- SD Card Tools (e.g. Belena Etcher, SD Card Formatter, RPI Imager)
- ROS2 Full Desktop
- Python
- OpenCV
- Docker-Desktop and Docker
- Fusion360 or Inventor 2026/2027
- PDF Viewer

Any changes or updates to this list will be posted on the WorldSkills Discussion Forum. Experts can propose changes in the Discussion Forum, and if approved, they will be available to all Competitors. Competitors who wish to have localization in a different language must apply at least C-3 months in advance. This is a hard deadline, otherwise they will receive an English OS. Note that support for other languages is not available.

4.3 Equipment and material not permitted

- Any additional laptop or portable computers.
- PDA, e.g. Palm, IPAQ, etc.
- Memory Sticks/MP3 Players/Digital Storage (Except on C-2 when installing software and loading code)
- Walkman Radio/CD Player
- Electronic Organizer/Diaries
- Wireless communications devices. (Phones can only be used during lunch break outside the workshop)
- Any additional software not supplied by the Competition Organizer, unless approved by Experts on C-2.
- Purchased items modified in any way before the Competition.
- Equipment that is similar or operates similarly to the supplied equipment. For example, if a metal cut-off saw is provided in the workshop, Competitors may not bring their own. Note that manual and powered hand tools have been added to this non-permittance per the TD (8.5). There is only one exception in 8.3 of the TD. (Competitors may bring **two** electric screwdrivers)
- All raw materials, components, parts, purchased items, tools and equipment not permitted in the Technical Description and List of Materials Allowed and Not Allowed of the Skill.

The Skill Management Team could decide that additional items are not permitted before or during the competition, provided that the information is provided to Competitors and Experts.

4.4 Items to be provided by the Competitor

Each Competitor must provide and wear identification indicating their country or region. These must always be worn during the competition and should follow overall WorldSkills regulations.

Competitors may manufacture and create custom components for their robots. However, some restrictions will be enforced and checked before the Competitors can build their robots on C1. Checks will happen on C-2.

- (a) All 3D-Printed elements must be created using ABS, ASA, PLA, PETG, TPU, PC, or CF. If threaded inserts are used in 3D parts, their weight is included in the 3D printing cost.
- (b) All sheet components must be created using any polycarbonate or self-reinforced polypropylene.
- (c) Custom PCBs are permitted only for wiring purposes and fall under the same cost as sheet components.
- (d) The max budget may not exceed \$1000 points. The budget covers the cost of extra components from Studica, 3D-printed components, and Sheet components. Competitors' spare parts are not part of the budget. But they are included if used on the robot. If there are variations of components, they will all be weighed and count towards the budget.
- (e) Competitors must have at least 3 cloned copies of their SD card for the OS of the VMX. These copies are held by the SCM until/if needed.
- (f) Competitors must provide a USB with any software or code they wish to load onto the computers on C-2. The SCM will hold the USB for the rest of the competition.
- (g) Competitors must bring any components not included in the full collection with them to WorldSkills 2026 Shanghai. It will not be provided for them. This includes any budget items related to Studica parts.
- (h) On C-2, experts will approve of all items in 4.4 (A)(B)(C)(G). Items not approved will need to be modified and approved before installing on the robot.

- (i) Any personal protective equipment that the Competitor deems fit, outside of what is provided by the competition.

Additionally, Competitors may bring jigs, fixtures, and clamps for assembly. These items are not included in the budget (TD (9.2)) but will be held by the SCM when not in use. If the items become part of the robot, they will be counted towards the budget. Otherwise, they must be removed from the robot.

During the competition, no tools, equipment, stationery, components, parts, raw materials, manuals, drawings, electrical devices, or digital storage devices may be removed from or brought into the competition venue, unless approved by the Skill Management Team (SMT). The SMT must approve any additional items entered in the competition after Familiarization Day on C-2, and if approved, they must be checked immediately by the responsible Experts.

5. Main Pre-Test Project

The main Test Project is divided into various modules in accordance with the WorldSkills Occupation Standards (WSOS).

The WorldSkills Shanghai 2026 Autonomous Mobile Robotics Test Project requires Competitors to design, build, operate, and program a mobile robot capable of addressing ALL the performance requirements.

Building on the success of WorldSkills Lyon 2024, the Autonomous Mobile Robotics skill competition will combine and build upon the modules from Lyon. These modules focus on the individual and teamwork skills of the Autonomous Mobile Robotics Technician/Engineer.

5.1 Module A – Work Organization, management and communication

The first module addresses sections 1 and 2 of the WSOS: “Work Organization and Management” and “Communication and Interpersonal Skills.”

5.1.1 Completion Time

Module A – Work Organization, management and communication is observed throughout the competition.

5.1.2 Assessment Criteria

Work organization and management are important to the regular tasks that Competitors are expected to perform in a cooperative environment. Moreover, there are multiple ways for a Competitor to communicate, and it is expected that they will always do so in a professional manner.

Assessment is also part of each evaluation for other modules and is marked by experts overseeing that module.

Potential examples of criteria:

- Adherence to schedules.
- Co-operative behaviors with compatriot Competitors, opponent Competitors and the Experts.
- Workplace organization.
- Robot assembly completion time.
- Asking before leaving the Competitor’s work area.

- Robot communication lights/buzzer correctly demonstrate whether a robot is moving, stationary or has an error.

5.1.3 Module Task

There is no specific task relating to this module.

5.2 Module B – Design

Module B addresses section 3 of the WSOS: “Design”.

5.2.1 Completion time

Competitors have until lunch of C1 to complete the assembly of their robot. Competitors that are building robot OMSs or any part of the robot that is not part of the base robot may only attach it to the robot after lunch. Competitors who did not complete building the base robot before lunch will have the remainder of C1 to finish it.

5.2.2 Assessment Criteria

Competitors are assessed on the assembly and wiring of the base robot rolled by the dice during the morning of C1.

Marking for this module is done on C1 regardless of the Competitor's completion.

5.2.3 Module task

Competitors are required to assemble the robot base assigned from the dice roll at the C1 morning briefing. The robot OMS or extra parts can be assembled during C-2, but may not be attached to the robot until after lunch C1.

5.3 Module C – Prototyping, Robot Environment Interaction

Module C addresses sections 4 and 7 of the WSOS: “Prototyping” and “Robot Environment Interaction”.

5.3.1 Completion Time

Module C can be completed during one of the blocks on C2 – C4, depending on Competitor rotation.

5.3.2 Assessment Criteria

Competitors will be assessed based on the documentation provided during the module briefing.

5.3.3 Module Task

Competitors will be given a kit containing parts they will use with their robot. The kit could be unknown or known parts. Competitors will be required to complete the task outlined in the module briefing.

5.4 Module D – Navigation and Localization

Module D addresses section 5 of the WSOS: “Navigation and Localization”.

5.4.1 Completion time

Navigation and localization can be completed during one of the blocks on C2 – C4, depending on Competitor rotation.

5.4.2 Assessment Criteria

Competitors are given one run at the task outlined in the briefing. The module uses a purpose-built court.

5.4.3 Module task

Competitors will be given a ROS2 map of the navigation court. Competitors will be required to create points on the map as outlined by the task in the module briefing. Competitors will roll the dice to choose the order in which the robot must travel to each point. An example of this process is in Appendix A.

5.4.4 Equipment

Competitors will use their base robot built on C1.

5.5 Module E – Vision

Module E addresses section 6 of WSOS: “Vision”.

5.5.1 Completion Time

Vision can be completed during one of the blocks on C2 – C4, depending on the Competitor rotation.

5.5.2 Assessment Criteria

Module uses a purpose-built vision court mimicking the performance court.

5.5.3 Module Task

Competitors will complete the vision task outlined in the module briefing. A roll of the dice will indicate the correct task to complete during vision.

5.5.4 Equipment

Competitors will use their base robot built on C1. Cameras are mounted on the robot via the provided mount or on the Competitor's OMS.

5.6 Module F – Testing and Fault Finding

Module F addresses section 8 of WSOS: “Testing and Fault Finding”

5.6.1 Completion Time

Module F will be completed during one of the blocks on C2 – C4, depending on the Competitor rotation.

5.6.2 Assessment Criteria

Module F is assessed in a closed room.

5.6.3 Module Task

Competitors will be assigned a support problem issued by a fictitious client. The robot is malfunctioning, and the Competitors will be required to identify the problem via a checklist. Competitors may be required to perform repair and functional testing.

5.6.4 Equipment

Competitors will be provided with an Autonomous mobile robot that needs a service. A service checklist will be provided, and Competitors must follow the checklist. All tools required to diagnose and fix any issue(s) with the robot will be provided.

5.7 Module G – Commissioning

Module G addresses mainly section 9 of WSOS: “Performance and Commissioning”. Module G is unique in that it includes aspects of sections 1–8.

5.7.1 Completion time

Module G will be completed in two separate blocks on C2 – C4, depending on the Competitor rotation.

NOTE: The blocks are not back-to-back.

5.7.2 Assessment Criteria

There are two separate task runs to be completed.

If a robot requires Competitor interaction after the evaluation run has started (unless otherwise stated in the module task), such as using teleoperation or physical interaction to reorient the robot before resuming the autonomous routine, the evaluation run can continue, but will not receive full marks.

Competitors may interact with the robot in this situation multiple times during an evaluation, BUT MUST NOTIFY AN EXPERT BEFORE DOING SO.

5.7.3 Module task

Example tasks are located in Appendix B.

5.7.4 Equipment

Competitors will use their robot for the module.

5.7.5 Objects

Object details are specified in Appendix C, which highlights every required detail. Competitors will only find out what the object(s) are on C1. The exact object(s) will not be disclosed to anyone before C1.

5.7.6 Robot Specifications

The base collection can build 4 different robot drive bases and storage compartments required for completing all modules. Images of the base robots are demonstrated in Appendix.

1. Wheel Differential Drive
2. Mecanum Drive
3. X Drive
4. Four Wheel Differential All-Terrain Drive

Every Competitor will build the same drivetrain during C1 based on the morning dice roll. After Competitor robot evaluations for Module B are complete, Competitors may add an OMS if they have one. **Robot bases are not to be modified in any way from the original design during the competition. The only exception is for adding sensors.**

Recommended Requirements for Robot OMS:

- Robot gripper that can pick up the object. The gripper must have 2 – 3 degrees of freedom.
- Mechanism for changing the height/depth of the gripper from the robot's main frame.
- Camera mount for object detection.

- The robot OMS must be able to reach a height of above 500 mm. i.e., the largest platform height is 500 mm. If an object is 100 mm, a good OMS could reach 650 mm in height to ensure it can be placed properly at the drop-off location.
- The OMS must be able to extend to a maximum of 400 mm outside the robot base frame. Meaning that if the robot base was against a stand, the object could be up to 400 mm away from the edge of the stand.
- OMSs must be able to reach a minimum height of 100 mm. **No object will be placed below 100mm.**
- OMSs should have at least 180 degrees of rotation or more to ensure they can pick up on both sides of the robot, plus the back or front of the robot.
- An OMS may only hold one object at a time. If a robot wishes to hold more than one object, a separate storage system must be developed.

5.7.7 Court

No court will be given for any Pre-TPs. Competitors can use the object(s) specifications (5.7.5), drive bases (5.7.6), and previous Test Projects, such as WorldSkills Lyon 2024, to design their own court for practice and preparation.

5.8 Sample schedule

9:00 - 12:00		14:00 - 16:00		9:00 - 12:00		13:30 - 16:30		9:00 - 12:00		13:30 - 16:30		9:00 - 12:00		13:30 - 16:30	
C1 - Morning		C1 - Afternoon		C2 - Morning		C2 - Afternoon		C3 - Morning		C3 - Afternoon		C4 - Morning		C4 - Afternoon	
Team 1	Robot Base Building	Team 1	Attach OMS to base & other	Team 1	Vision	Free	Vision	Team 17	Vision	Team 13	Vision	Team 9	Vision	Team 5	Vision
Team 2		Team 2		Free		Team 18		Team 14		Team 10		Team 6			
Team 3		Team 3		Free		Team 19		Team 15		Team 11		Team 7			
Team 4		Team 4		Free		Team 20		Team 16		Team 12		Team 8			
Team 5	Robot Base Building	Team 5	Attach OMS to base & other	Team 5	Proto-typing	Team 1	Proto-typing	Free	Proto-typing	Team 17	Proto-typing	Team 13	Proto-typing	Team 9	Proto-typing
Team 6		Team 6		Team 2		Free		Team 18		Team 14		Team 10			
Team 7		Team 7		Team 3		Free		Team 19		Team 15		Team 11			
Team 8		Team 8		Team 4		Free		Team 20		Team 16		Team 12			
Team 9	Robot Base Building	Team 9	Attach OMS to base & other	Team 9	Com	Team 5	Com	Team 1	Com	Free	Com	Team 17	Com	Team 13	Com
Team 10		Team 10		Team 6		Free		Team 18		Team 14		Team 10			
Team 11		Team 11		Team 7		Free		Team 19		Team 15		Team 11			
Team 12		Team 12		Team 8		Free		Team 20		Team 16		Team 12			
Team 13	Robot Base Building	Team 13	Attach OMS to base & other	Team 13	N&L	Team 9	N&L	Team 5	N&L	Team 1	N&L	Free	N&L	Team 17	N&L
Team 14		Team 14		Team 10		Free		Team 18		Team 14		Team 10			
Team 15		Team 15		Team 11		Free		Team 19		Team 15		Team 11			
Team 16		Team 16		Team 12		Free		Team 20		Team 16		Team 12			
Team 17	Robot Base Building	Team 17	Attach OMS to base & other	Team 17	T&F	Team 13	T&F	Team 9	T&F	Team 5	T&F	Team 1	T&F	Free	T&F
Team 18		Team 18		Team 14		Free		Team 18		Team 14		Team 10			
Team 19		Team 19		Team 15		Free		Team 19		Team 15		Team 11			
Team 20		Team 20		Team 16		Free		Team 20		Team 16		Team 12			
				Free	Com	Team 17	Com	Team 13	Com	Team 9	Com	Team 5	Com	Team 1	Com
				Free		Team 18		Team 14		Team 10		Team 6		Team 2	
				Free		Team 19		Team 15		Team 11		Team 7		Team 3	
				Free		Team 20		Team 16		Team 12		Team 8		Team 4	
Morning - 3 Hrs working time followed by 1 hr evaluation				3 Hrs working time followed by 1 hr evaluation				3 Hrs working time followed by 1 hr evaluation				3 Hrs working time followed by 1 hr evaluation			
Afternoon - 2 Hrs working time															

This is a sample schedule to explain better and demonstrate the rotation system. In this schedule, up to 24 teams can be accommodated in a 4-day event.

Appendix A – Module F Navigation

The flow for navigation will be as follows:

- Competitors receive the map of the navigation area. An example is given in Figure 1.0.
- Competitors will roll the dice and are given the point locations on the map. An example is given in Figure 1.1.
- Competitors roll the dice again and are given the order the robot will have to drive between the points.
 - Example Roll 1: $A > B > D > A > C > B$
 - Example Roll 2: $B > C > A > D > B > A$
- During the evaluation, Competitors will place the robot on the starting point.
- Experts will place a block to close one of the passageways.
- Competitors will hit start on the robot, and the run will start.

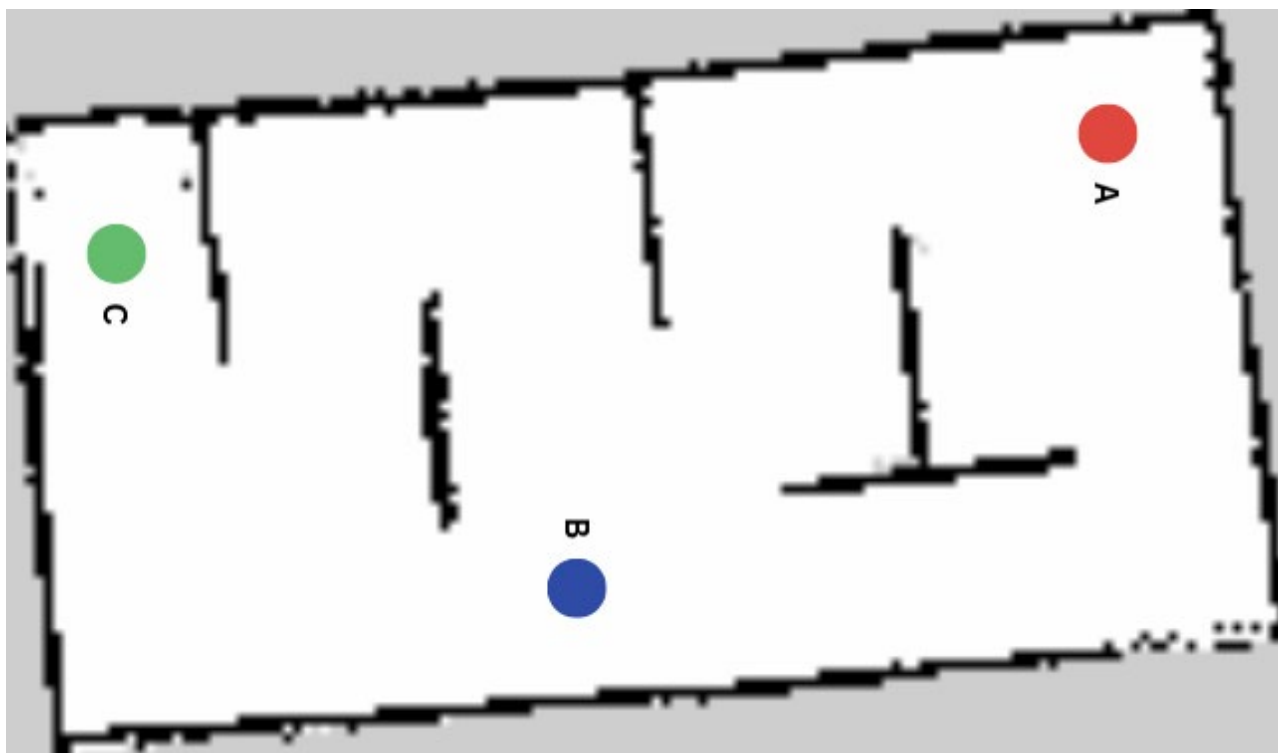
Figure 1.0 – Demo Map

This is an example map of a 2 m x 4 m court area.



Figure 1.1 – Demo Map with Points

Same example map, but points have been labelled.



Appendix B - Module G – Commissioning

The task might involve a client who needs to collect and deliver various components within an environment. The number of items to be collected or delivered will be unknown. In this environment, there might be three pickup locations. The robot will be required to scan these pickup locations for objects. Depending on the object, the robot will pick up the object(s) and deliver them with precision and accuracy to the drop-off location for that object. The drop-off locations will be determined with a roll of the dice at the start of the module. However, the objects at the pickup locations are determined at the start of the evaluation by rolling a die. When there are no more objects in the pickup locations, the robot may return to the start area and finish the run.

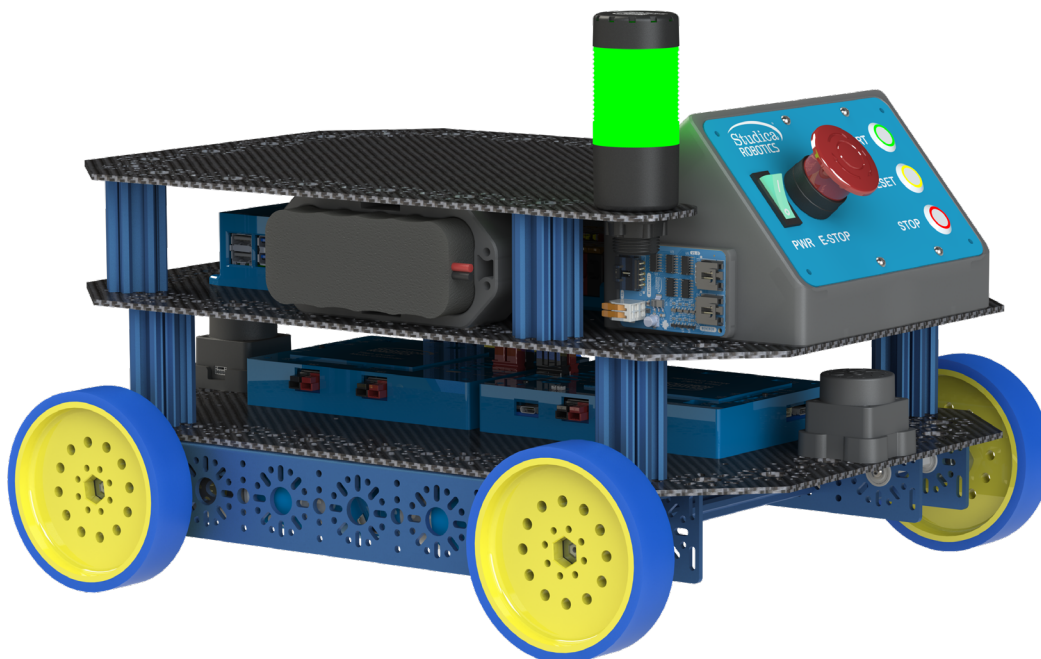
Example Run Sequence:

- At the start of the module, a die is rolled to determine drop-off locations and the objects that match that location.
- During the assessment, the robot is placed in the start area.
- Another roll of the dice is taken to determine which objects are placed in the pickup locations.
- Competitor starts the run.
- Robot scans the pickup locations for the objects, picks up the objects, and delivers them to the drop-off locations.
- Competitors replace the objects in the pickup locations until the objects run out.
 - Example: three of object A, 4 of object B, 2 of object C
- Robot repeats until the pickup locations are empty and returns to the start area.
- At the start area, the robot will flash the LED tower RED to indicate the run has ended.

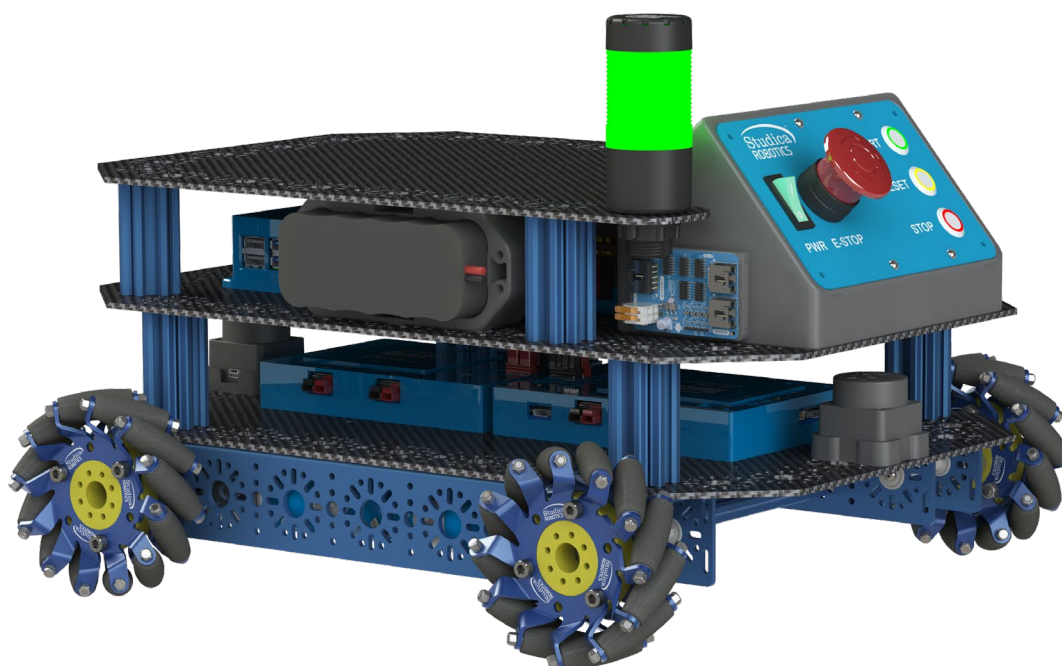
Appendix C – Robots and Objects

Base Robots

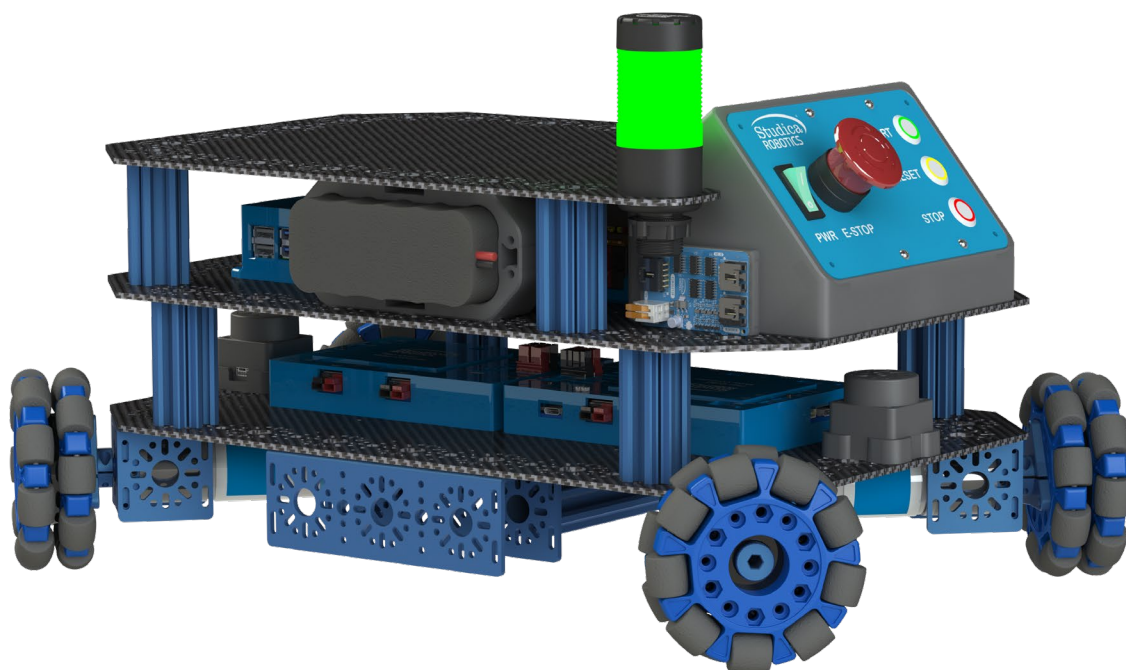
4 Wheel Differential Drive



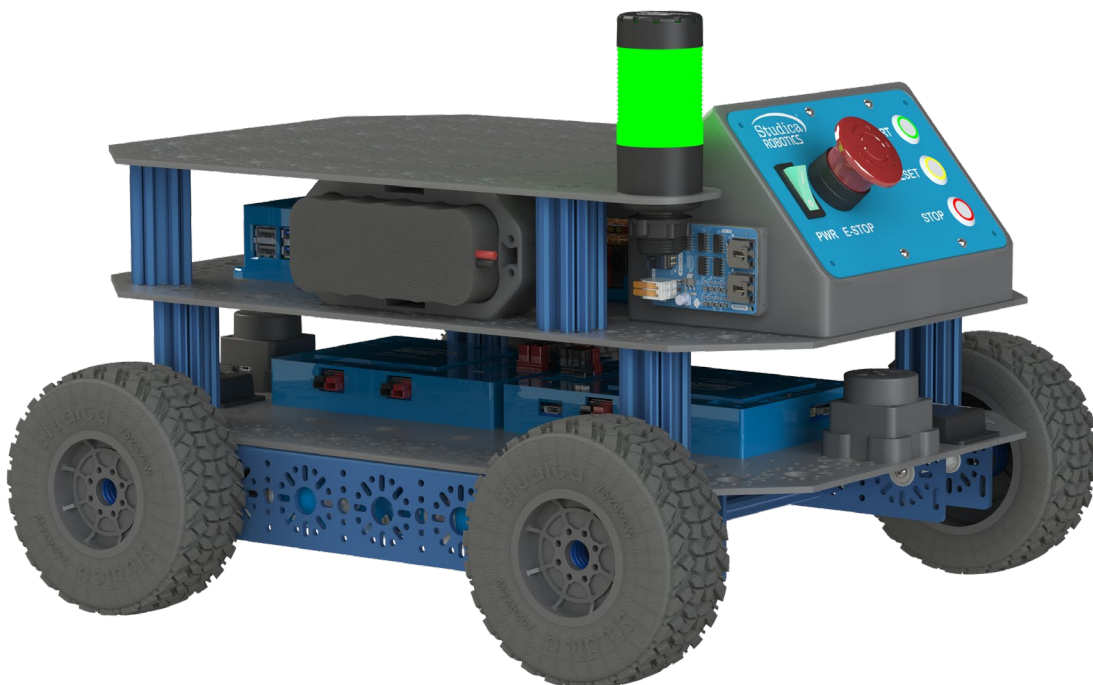
Mecanum Drive



X Drive



4 Wheel Differential Drive (All-Terrain)



Objects

There may be a total of four objects, each with multiple colours, surfaces, and labels.

Object #1

- Cylindrical
- Metallic
- Width: 50 mm – 75 mm
- Height: 100 mm – 160 mm
- Weight: 250 g – 400 g

Object #2

- Rectangular
- Plastic and smooth
- Width: 80 mm – 110 mm
- Length: 120 mm – 180 mm
- Height: 40 mm – 60 mm
- Weight: 200 g – 300 g

Object # 3

- Semi - Spherical
- Wood or plastic
- Width: 120 mm – 160 mm
- Height: 50 mm – 70 mm
- Weight: 50 g – 500 g

Object #4

- Mystery released on C1.